

Python Practice Handout

Input

1. Asking the user a question

Input lets the person running your program type something in. Write input with a string inside the brackets. That string is the prompt shown in the console.

```
input("Enter some text")
```

Run it and the console waits for you to type. Whatever you type disappears though, because you have not kept it anywhere.

2. Keeping what they typed

Store the input in a variable, then you can use it. Here whatever is typed goes into `user_text`.

```
user_text = input("Enter some text: ")  
print(user_text)
```

Console:

```
Enter some text: Hello  
Hello
```

The first Hello is what you typed. The second is what the program printed back.

3. A space after the prompt

Without a colon and a space at the end of the prompt, your typing runs straight into the last word: `Enter some textHello`. Ending the prompt with a colon and a space gives `Enter some text: Hello`, which is far easier to read. Notice the space inside the quotes in the examples above.

4. Using what they typed

Once it is in a variable you can do anything with it, such as putting it in capitals.

```
user_text = input("Enter some text: ")  
print(user_text.upper())
```

Console:

```
Enter some text: Hello  
HELLO
```

5. Input is always a string

This is the part that catches everybody. Whatever the user types comes back as a string, even when it looks like a number.

Multiplying a string by 2 repeats it, so typing 8 gives 88 rather than 16.

```
user_number = input("What do you want to double? ")  
  
print(user_number * 2)
```

Console:

```
What do you want to double? 8  
88
```

6. Turning it into a number

Wrap the variable in int to convert the string into a whole number. Then the maths works as expected.

```
user_number = input("What do you want to double? ")  
  
print(int(user_number) * 2)
```

Console:

```
What do you want to double? 8  
16
```

If you need a decimal instead, use float in the same way.

7. Try it yourself

Ask the user for some text, then ask whether they want it in capitals or in small letters, and print it their way. Try this on your own before looking below.

```
user_text = input("Enter some text: ")  
upper_or_lower = input("Type 1 for upper, 2 for lower: ")  
  
if upper_or_lower == "1":  
    print(user_text.upper())  
else:  
    print(user_text.lower())
```

Console, typing Nick then 1:

```
Enter some text: Nick  
Type 1 for upper, 2 for lower: 1  
NICK
```

Console, typing Nick then 2:

```
Enter some text: Nick  
Type 1 for upper, 2 for lower: 2  
nick
```

Look closely at the if statement. It compares with "1" in quotes, not the number 1, because input always hands back a string. Without the quotes it would never match. Anything other than 1 falls to the else and gives small letters.